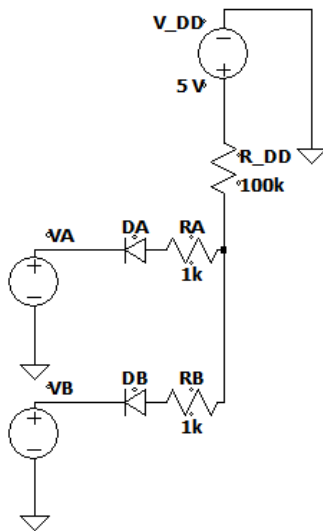


Basic Operation

Exercise 1



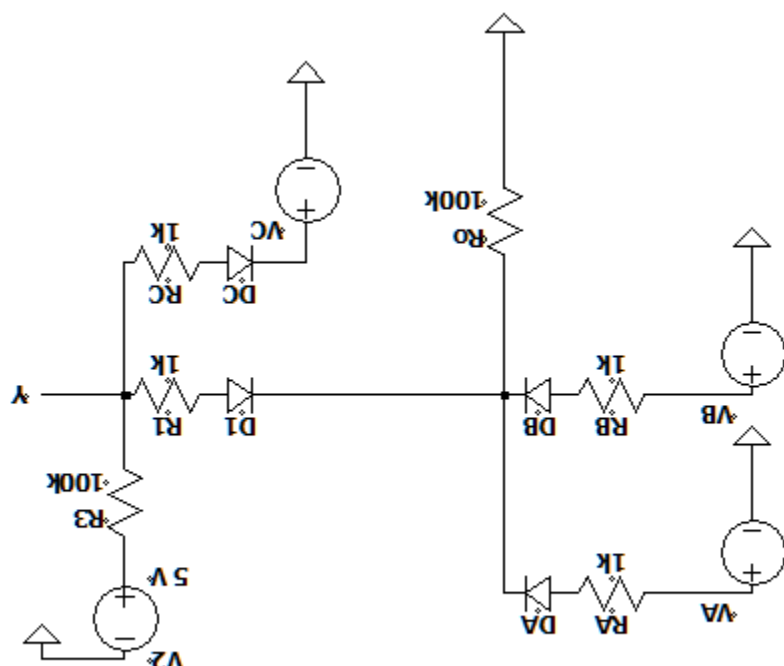
For the AND gate-

- For logic case (1,0), find the output voltage & verify your assumption of the diode model used.
- Find the current through R_{DD} for logic case (0,1).
- If R_B is doubled, find the currents through the diodes for logic case (0,0).
- Find the voltage of the node between D_B & R_B for the case in (c)

Ans: a) 0.743 V
b) 0.043 mA
c) 0.028 mA, 0.014 mA
d) 0.7 V

Practice Problem 1:

Implement the Boolean function, $Y = (A+B)C$ using diode logic.



Ans:

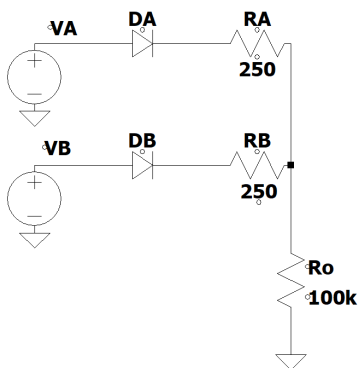
Exercise 2

Find the lower threshold voltage for the AND gate in **Exercise 1**, so that the output logic state can be determined correctly.

Ans: 0.743 V

Power Dissipation

Exercise 3



Find the power dissipation for all input logic cases.

[High input = 5 V, Low input = 0.2 V]

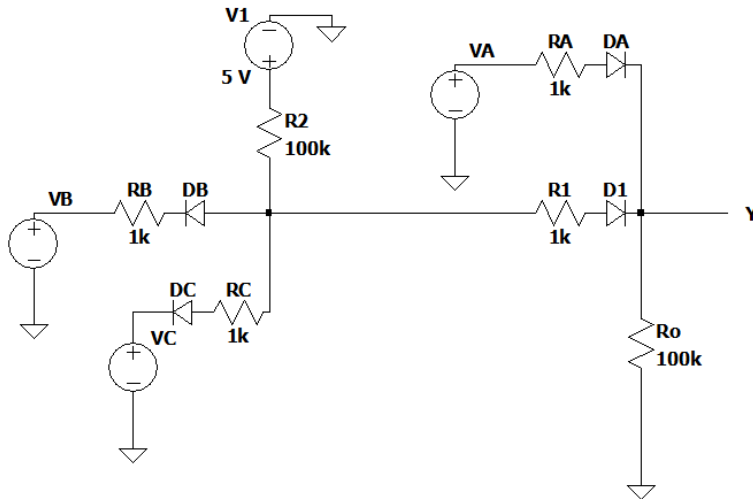
Ans: 0 mW, 2.145 μ W, 214.5 μ W, 214.5 μ W

Practice Problem 2:

Find the maximum & average power dissipation for the OR gate in **Exercise 3**.

Ans: 21.45 μ W, 16.0875 μ W

Exercise 4



For the diode logic circuit given-

- Find the Boolean expression of Y.
- Determine the higher & lower threshold of output voltage.
- Find the maximum & average power dissipation of the full circuit.

Ans:

a) $Y = A + BC$

b) 2.14 V, 0 V

c) 3.8 mW, 0.742 mW

For the circuit in Figure 4, assume **Logical High** is $5V$ and **Logical Low** is $0V$.

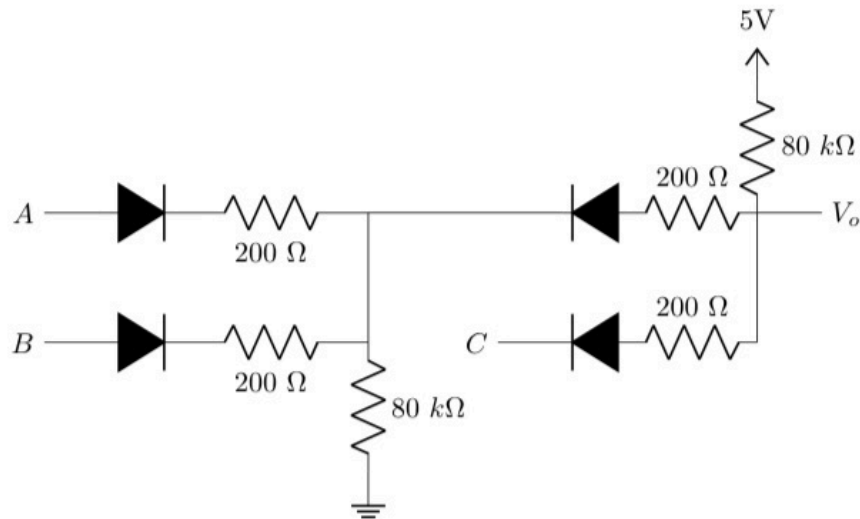


Figure 4

CO1	(a)	Determine the logic function this circuit is implementing.	[2]
CO1	(a)	Calculate the output voltage and power dissipation of the entire circuit when A=Low, B=High, C=Low.	[8]

b) $(A + B)C$

c) KCL at output, $\frac{5 - V_o}{80} = \frac{V_o - 0.7}{0.2} \rightarrow V_o = 0.71 \text{ V}$

$$P = (5 - 0) \times \frac{5 - 0.7 - 0}{0.2 + 80} + (5 - 0) \times \frac{5 - 0.71}{80} = 0.536$$